

The prevalence and burden of generalized anxiety disorder in the United States healthcare system: Real-world prevalence and incidence from 2020 to 2023

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ABSTRACT

Background: Generalized anxiety disorder (GAD) is characterized by persistent, difficult-to-control worry that is often accompanied by substantial comorbidity, impairment, and reduced quality of life. Yet, its prevalence in the United States remains uncertain and may be underestimated.

Methods: We estimated the prevalence and incidence of diagnosed GAD among adults in the U.S. healthcare system using payer-complete closed claims data from the Komodo Healthcare Map™ database from January 1, 2020 to December 31, 2023. Patients with GAD were identified using ICD-10 code F41.1. Annual and three-year prevalence, as well as annual incidence rates, were calculated and projected to the U.S. adult population.

Results: From 2021–2023, most patients diagnosed with GAD were women, middle-aged, and commercially insured with women diagnosed at more than twice the rate of men. Projected annual prevalence increased from 5.4% in 2020–6.6% in 2023, with a 3-year prevalence of 10.3% over the 2021–2023 period. The projected annual incidence ranged from 2.1% to 2.3%.

Conclusion: The increasing prevalence of GAD diagnoses may reflect multiple converging factors, including increased screening and recognition. Together, these updated national estimates highlight not only the increasing prevalence of GAD but also the need to expand access to care and improve the effectiveness of treatments.

1. Introduction

Among anxiety disorders, generalized anxiety disorder (GAD) is characterized by persistent, difficult-to-control anxiety, accompanied by dyssomnia, impaired concentration, irritability, and muscle tension [1]. GAD is often accompanied by a range of psychiatric and physical conditions that add to its overall disease burden, including frequent pain [2, 3], sleep disorders [4] and other non-psychiatric comorbidities (e.g., migraines, irritable bowel syndrome, etc.) [5]. Further, GAD is associated with substance use disorder and major depressive disorder (MDD), with MDD observed in approximately 50%–70% of patients with GAD symptoms [6–10]. GAD is also associated with significant functional impairment [11–13], which worsens with increasing GAD severity [12]

and is comparable in magnitude to the impairments seen in MDD [14–16]. Together these factors contribute to significant direct and indirect healthcare costs and increased healthcare resource utilization [11,17,18], which produces a substantial economic burden for both patients and the healthcare system.

Despite its clinical and societal impact, the prevalence of GAD in the United States (U.S.) remains difficult to estimate and is likely underestimated. Many estimates are outdated, based on *DSM-IV* diagnostic criteria, or derived from survey methods with methodological limitations. The most widely cited diagnostic interview-based estimates come from the National Comorbidity Survey Replication (NCS-R), which used in-person, structured interviews (World Mental Health version of the WHO Composite International Diagnostic Interview [CIDI]) in the early

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2000s. The NCS-R reported a 12-month prevalence of 2.7% and a lifetime prevalence of 5.7%, with higher rates among women than men [19].

More recent federal surveillance efforts have relied on symptom-based screening instruments rather than structured diagnostic assessments. For example, the National Health Interview Survey (NHIS) used the Generalized Anxiety Disorder-7 (GAD-7) [20] to assess anxiety symptoms over the prior two weeks and found the highest symptom burden among adults aged 18–29 years, women, individuals with less than a high school education, those with household incomes below 100% of the federal poverty level, and residents of rural areas [20–22]. However, because the GAD-7 captures symptoms over a two-week window [20], scores may be influenced by transient stressors. In addition, symptom overlap with related conditions (e.g., insomnia in both depressive and anxiety disorders, restlessness in both GAD and major depression) may bias prevalence estimates. In contrast, stigma and recall bias may contribute to underreporting. Although these surveys provide important population-level data, cross-sectional, symptom-based studies estimate the prevalence of anxiety symptoms rather than the prevalence of clinically-diagnosed GAD.

Despite the significant burden of illness, GAD occurring outside specialty psychiatric settings has received comparatively little attention, particularly with regard to its co-occurrence with non-psychiatric medical conditions. As a result, the extant prevalence estimates may substantially underestimate the true burden of GAD and contribute to suboptimal management and healthcare resource allocation. Updated real-world prevalence estimates, particularly in individuals that are actively engaged with the healthcare system, are needed. Integrating claims data with recent epidemiologic findings provides an opportunity to refine estimates of GAD prevalence and incidence, its stability over time, and its relationship to co-morbid conditions. With these considerations in mind, we sought to estimate the prevalence and incidence of GAD among adults in the U.S. engaging with the healthcare system using claims data from 2020 to 2023.

2. Methods

2.1. Study design and data sources

This retrospective, longitudinal study identified adults with GAD using International Classification of Diseases, Tenth Revisions (ICD-10) diagnosis codes for GAD between January 1, 2020 to December 31, 2023, the most recent, complete years available. The analysis used de-identified, Health Insurance Portability and Accountability Act (HIPAA)-compliant patient-level data from the Komodo Healthcare Map™ claims database. This large dataset is broadly representative of the U.S. population and includes both open claims data and adjudicated payer-complete medical and pharmacy claims.

2.1.1. Patient inclusion/exclusion criteria

Patients with GAD were identified using ICD-10 diagnosis code F41.1. Inclusion in the study required either two or more medical claims at least 30 days apart containing ICD-10 code F41.1 in the study time period or one medical claim containing ICD-10 code F41.1 and one or more GAD pharmacotherapy treatments (Table A.2) in the study time period (Table 1).

Patients were excluded if they were under the age of 18 or had missing data on age or gender. Patients were also excluded if they had another neuropsychiatric condition (Table A.3) and only one diagnosis F41.1 GAD code. A sensitivity analysis was conducted to explore the impact of removing this exclusion criterion.

The utilization of claims data for this analysis is consistent with best practices for studies conducted using electronic health care data and studies with a longitudinal assessment [23,24]. Particularly, the methodology as described above, with a two-year lookback period, is appropriate for confirming GAD cases [24].

Table 1
Cohort Inclusion and Exclusion Criteria.

Epidemiology Cohort Criteria – Projected Prevalence and Annual Incident			
Universal Inclusion Criteria	• 2 or more claims with diagnosis code F41.1, at least 30 days apart OR • 1 claim with diagnosis code F41.1 AND 1 + GAD-related treatment claim without comorbid neuropsychiatric conditions (See Table A.3: Other Neuropsychiatric Conditions for complete list)		
Prevalent Cohort	Time Period	Exclusion Criteria	Continuous Enrollment
Incident Cohort	• 1-year intervals from 2020 to 2023	• Missing age or sex	Require 12 months of continuous enrollment during the latest calendar year of the prevalence time-period
	• 3-year interval 2021–2023	• < 18 years of age at diagnosis	Require continuous enrollment in the respective time frame and 2-calendar years pre-index
Incident Cohort	• 1-year intervals from 2020 to 2023	• Incident – GAD Diagnosis 2 years prior to Index date (clean-period)	

2.1.2. Prevalence and incidence estimation

Payer-complete closed claims data were used to estimate annual and multi-year prevalence and annual incidence trends. Prevalence was calculated in single year intervals from 2020 through 2023. Diagnosis rates observed in the Komodo sample were projected to the U.S. population (see Fig. 1 for methodologic overview and Table A.1 for U.S. census data). For single-year prevalence analyses, diagnosis rates were calculated using the number of patients diagnosed with GAD as the numerator and all patients continuously enrolled during the corresponding calendar year as the denominator; for multi-year prevalence analyses, the denominator consisted of patients continuously enrolled in the most recent calendar year. Diagnosis rates were stratified by age, gender (as defined in the Komodo database), and payer channel. Stratified rates were then applied to corresponding U.S. Census population estimates to generate national projections [25].

Incidence estimates were calculated in one-year intervals from 2020 to 2023 with diagnosis rates in the Komodo sample scaled to the U.S. population. Incident case counts were calculated by applying the estimated incidence rate to the projected U.S. GAD prevalence [26]. For incidence analyses, the denominator included patients with continuous enrollment during the index year and the two preceding calendar years, including those with prevalent GAD. The numerator consisted of patients with a new GAD diagnosis in the index year and no documented GAD diagnosis during the prior two calendar years.

3. Results

3.1. Patient demographics

Among patients identified with GAD over the 2021–2023 period (n = 25.3 M), most were female (67%), middle-aged (mean age 43.7 years), and commercially insured (61%) (Table 2). Between 2021 and 2023, most patients were 18–44 years old (54%), followed by those 45–64 years old (29%); only 2% were 85 years old or older. Demographic analyses across each 1-year time period reveal that females appeared in GAD claims data at more than twice the rate of males. Additionally, among those patients identified with GAD from 2021 to 2023, the most common baseline co-occurring conditions in a 1-year lookback period pre-index were MDD (30%), neck and back pain (19%), obesity (17%), abdominal pain (16%), chronic fatigue syndrome (15%), back pain (14%), breathing abnormalities (13%), chest pain (12%), and insomnia (11%).

3.2. GAD prevalence trends

Among adults in the U.S. engaging with the healthcare system, the

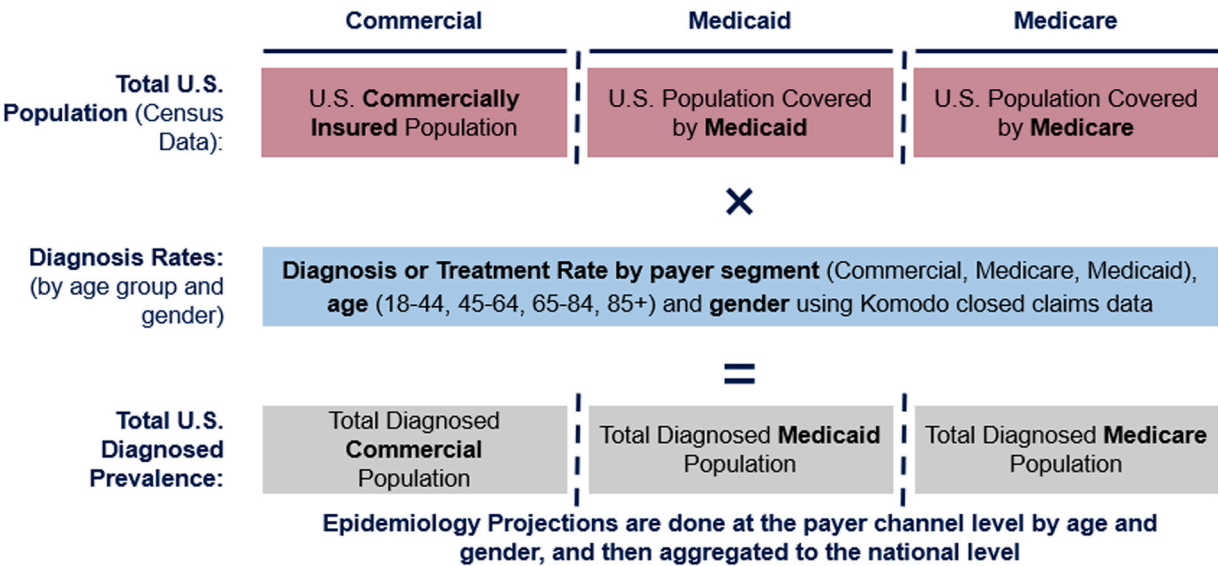


Fig. 1. Projection Methodology for Prevalence Estimate, Methodology to calculate the projected total U.S. diagnosed prevalence of GAD, which utilizes the total U.S. population from Census Data as well as diagnosis/treatment rates by payer segment, age, and gender. Abbreviations: U.S.: United States.

Table 2
Demographics of Patients with GAD.

Timeframe	1-Year Time Period				3-Year Time Period
Demographic	2020 <i>N</i> = 13.3 M	2021 <i>N</i> = 14.6 M	2022 <i>N</i> = 15.4 M	2023 <i>N</i> = 16.4 M	2021 – 2023 <i>N</i> = 25.3 M
Sex					
Female	68.5%	68.3%	68.2%	68.0%	67.4%
Male	31.5%	31.7%	31.8%	32.0%	32.6%
Age					
18–44	55.0%	53.6%	54.6%	54.5%	53.7%
45–64	30.0%	29.1%	28.7%	28.9%	29.4%
65–84	15.0%	15.4%	14.9%	14.8%	15.1%
85 +	0.0%	1.9%	1.8%	1.8%	1.8%
Mean (Age)	43.5	43.4	43.3	43.4	43.7
Median (Age)	42	41	41	41	42
Insurance type					
Commercial	62.7%	59.8%	61.5%	61.2%	60.6%
Medicaid	17.2%	17.9%	17.1%	17.5%	17.9%
Medicare	20.1%	22.3%	21.4%	21.3%	21.5%

projected annual prevalence of GAD increased from 5.4% in 2020–6.6% in 2023, based on patients meeting inclusion criteria (Section 2.1.1) of at least two GAD diagnoses or one GAD diagnosis accompanied by at least one GAD-related pharmacologic treatment (Fig. 2A). Over the

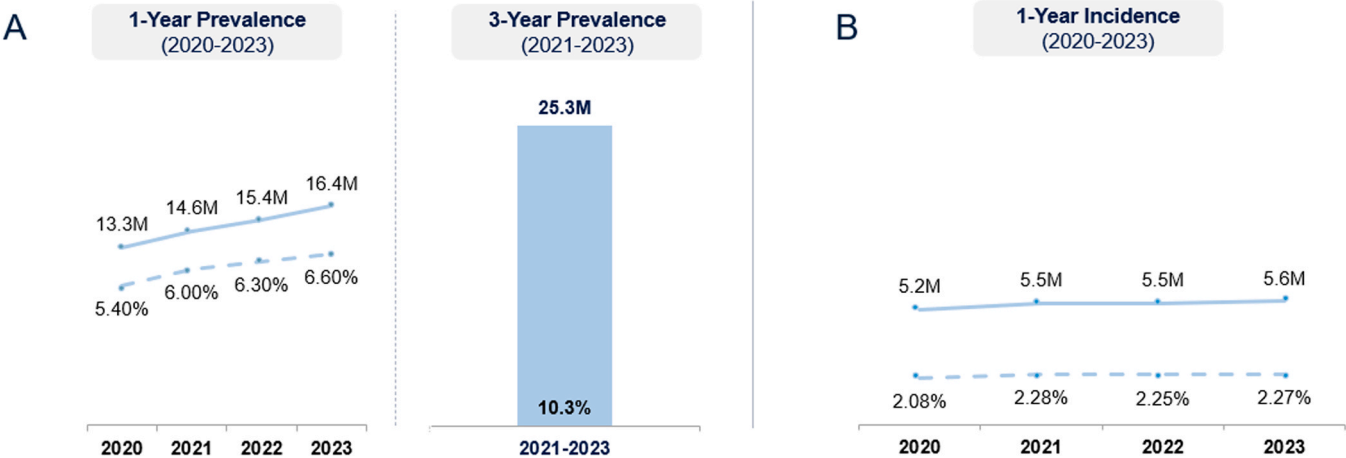


Fig. 2. Projected Prevalence and Incidence of GAD in U.S. Adults, A) Projected 1-year prevalence from 2020 to 2023, 3-year prevalence from 2021 to 2023. B) 1-year incidence from 2020 to 2023, Abbreviations: U.S.: United States.

2021–2023 period, the cumulative 3-year prevalence rate was 10.3%. Detailed projected prevalence estimates including U.S. Census figures are shown in **Table A.1**.

A sensitivity analysis was performed to assess the prevalence impact of inclusion/exclusion criteria on prevalence estimates. In the base case, patients with only a single GAD diagnosis and at least one related pharmacotherapy claim, who also had co-occurring neuropsychiatric conditions, were excluded to reduce the likelihood of GAD misclassification (e.g., medications prescribed for other psychiatric indications, such as major depressive disorder). When this exclusion criterion was removed, the estimated one-year prevalence of GAD in 2023 increased from 6.6% in the base case to 7.2%.

3.3. Annual incidence of GAD

The annual incidence of GAD in the U.S. among adults engaging with the healthcare system ranged from 2.1% to 2.3%, using a 2-year pre-index lookback to confirm no prior diagnosis and identify newly diagnosed cases (**Fig. 2B**). Detailed projected incident estimates including U.S. Census data and sample incident rates are available in **Table A.4**.

4. Discussion

This study provides contemporary, real-world estimates of the prevalence and incidence of GAD in the U.S. for adults engaging with the healthcare system and demonstrates a substantially greater utilization burden than previously reported. Importantly, more than one in ten adults met criteria for GAD over a three-year period, exceeding lifetime prevalence estimates from prior general population studies, which have ranged from approximately 5% to almost 8% [13,19,27]. In addition, the projected 1-year prevalence of GAD increased from 5.4% in 2020–6.6% in 2023. Consistent with prior epidemiologic studies, women, commercially insured adults, and middle-aged patients were disproportionately represented [19].

Several factors potentially contribute to this upward trend. These include a true increase in anxiety-related morbidity as well as improvements in recognition and screening [28] and diagnostic accuracy. Broader societal stressors (e.g., economic instability, social disruption related to the COVID-19 pandemic, environmental factors) may have further increased anxiety symptoms in the general population [29,30]. At the same time, evolving attitudes toward mental health, reduced stigma, and greater public awareness may have decreased barriers to seeking treatment. Together, these factors likely converge to increase both the incidence of GAD and the probability that affected individuals receive a formal diagnosis.

Despite improvement in detection and the 2023 U.S. Preventive Services Task Force (USPSTF) recommendation for routine anxiety screening in adults, GAD remains substantially underrecognized. Estimates suggest that up to half of individuals with GAD never receive a formal diagnosis [31–33]. While claims-based data analyses provide high specificity in identifying patients who are formally diagnosed, claims data only represent treatment-seeking patients, not the true overall GAD burden including undiagnosed individuals. Undiagnosed individuals have lower health related quality of life, greater healthcare resource utilization and economic burden and are differentially affected by other chronic psychiatric and noncommunicable diseases [17, 34–37]. Underestimation of GAD may contribute to gaps in health policy, efficiency in the allocation of healthcare resources, and clinical care pathways.

The increase in diagnosed GAD has implications for clinicians, health systems and policy makers. Although diagnosis can validate patient experiences and facilitate access to care, persistent limitations in treatment availability, access to specialty psychiatric care, high quality psychotherapy, and variable treatment effectiveness highlight the need for better, integrated models of care and innovation in pharmacotherapy treatments.

While this study leverages a large, representative claims database, there are several important limitations. First, this analysis utilized claims data which have inherent biases that should be considered. Claims data represent treatment-seeking patients, which may limit generalizability to a broader population. Additionally, by restricting the cohort to individuals with continuous data availability, there is the potential of introducing selection bias. Claims data are derived from diagnosis codes assigned at the point of care and therefore reliability of clinician-assigned diagnosis codes is important. In practice, clinicians are often administratively required to enter only a single diagnosis. For example, in a patient with generalized and social anxiety disorders and co-occurring panic disorder, a clinician might record only panic disorder. In some cases, coding decisions are influenced by insurance coverage (e.g., when a treatment is reimbursed only under a depressive disorder diagnosis, despite efficacy in both depressive and anxiety disorders). Thus, a clinician may capture the depressive disorder over the anxiety disorder. Diagnostic specificity in claims data is also limited. Clinicians may prefer coding “unspecified anxiety disorder,” which requires less diagnostic investigation and carries no implications for treatment authorization, rather than coding a more precise anxiety disorder. This coding behavior is compounded by the poor reliability of correct GAD diagnosis by clinicians [38], and measures of inter-rater or test-retest diagnostic reliability are not available from the claims data utilized for this analysis. Taken together, the claims data likely contain an underdiagnosed and misdiagnosed GAD population, and therefore this analyses may underestimate the true prevalence and incidence in the U.S. general population.

Second, until recently, routine screening in primary care has focused far more on depression (e.g., PHQ-9) than on anxiety. Because several core symptoms of GAD (e.g., sleep disturbance, poor concentration, fatigue, and psychomotor agitation) overlap with items on depression screeners, patients with primary anxiety disorder may have had these symptoms misclassified as depressive disorders. This highlights an important clinical pitfall: measurement-based care and screening tools are not substitutes for a full diagnostic assessment. Without deliberate evaluation for GAD, reliance on depression-only screeners may have inadvertently shifted diagnostic coding toward depressive disorders. Third, certain populations are underrepresented in this analysis. For instance, veterans receiving care in the VA Healthcare system were not included, despite evidence that anxiety disorders are more prevalent in this population and that veterans have higher rates of treatment resistance and psychiatric comorbidity compared with civilians [39]. Their exclusion likely attenuates severity estimates. Fourth, we required two-year continuous enrollment prior to index to establish incident cases. Although sensitivity analyses suggest that some patients have GAD-related claims beyond the two-year lookback period, extending the lookback requirement increased bias towards individuals with long-term, uninterrupted insurance coverage. Therefore, a longer lookback window was not adopted.

5. Conclusion

This study provides an updated real-world U.S. estimate of the prevalence and incidence of diagnosed GAD in adults. Rates observed among individuals engaged with the healthcare system were substantially higher than those reported in earlier epidemiologic studies [19], highlighting the need to re-evaluate the current treatment paradigm and additional studies of treatment utilization in GAD, as both psychotherapy and pharmacotherapy use remain low among patients with anxiety disorders, with utilization patterns influenced by comorbidity, sex (greater use among females) [40], and age [41]. Additionally, whether recent USPSTF recommendations for anxiety screening alter real-world prevalence estimates, diagnostic coding practices, and ultimately treatment engagement will need to be determined. While the unmet treatment needs in GAD are substantial, these updated real-world prevalence and incidence data suggest that the public health burden is

even greater than previously suggested, which highlights the need to expand access to care and improve treatment options for individuals with GAD.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.xjmad.2026.100172](https://doi.org/10.1016/j.xjmad.2026.100172).

References

- [1] Association A.P. *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition*. 2013.
- [2] Beesdo K, Hoyer J, Jacobi F, et al. Association between generalized anxiety levels and pain in a community sample: evidence for diagnostic specificity. *J Anxiety Disord* 2009;23(5):684–93. <https://doi.org/10.1016/j.janxdis.2009.02.007>.
- [3] De La Rosa JS, Brady BR, Ibrahim MM, et al. Co-occurrence of chronic pain and anxiety/depression symptoms in U.S. adults: prevalence, functional impacts, and opportunities. *Pain* 2024;165(3):666–73. <https://doi.org/10.1097/j.pain.0000000000003056>.
- [4] Marcks BA, Weisberg RB. Co-occurrence of insomnia and anxiety disorders: a review of the literature. *Am J Lifestyle Med* 2009;3(4):300–9. <https://doi.org/10.1177/1559827609334681>.
- [5] Sareen J, Jacobi F, Cox BJ, et al. Disability and poor quality of life associated with comorbid anxiety disorders and physical conditions. *Arch Intern Med* 2006;166(19):2109–16. <https://doi.org/10.1001/archinte.166.19.2109>.
- [6] Bittner A, Goodwin RD, Wittchen HU, et al. What characteristics of primary anxiety disorders predict subsequent major depressive disorder? *J Clin Psychiatry* 2004;65(5):618–26. <https://doi.org/10.4088/jcp.v65n0505>.
- [7] Kessler RC, Sampson NA, Berglund P, et al. Anxious and non-anxious major depressive disorder in the world health organization world mental health surveys. *Epidemiol Psychiatr Sci* 2015;24(3):210–26. <https://doi.org/10.1017/S2045796015000189>.
- [8] Zhou Y, Cao Z, Yang M, et al. Comorbid generalized anxiety disorder and its association with quality of life in patients with major depressive disorder. *Sci Rep* 2017;7(1):40511. <https://doi.org/10.1038/srep40511>.
- [9] Asselmann E, Wittchen HU, Lieb R, et al. Sociodemographic, clinical, and functional long-term outcomes in adolescents and young adults with mental disorders. *Acta Psychiatr Scand* 2018;137(1):6–17. <https://doi.org/10.1111/acps.12792>.
- [10] Behrendt S, Beesdo-Baum K, Zimmermann P, et al. The role of mental disorders in the risk and speed of transition to alcohol use disorders among community youth. *Psychol Med* 2011;41(5):1073–85. <https://doi.org/10.1017/s0033291710001418>.
- [11] Revicki DA, Travers K, Wyrwich KW, et al. Humanistic and economic burden of generalized anxiety disorder in North America and Europe. *J Affect Disord* 2012;140(2):103–12. <https://doi.org/10.1016/j.jad.2011.11.014>.
- [12] Duong P., Suponcic S.J., Finlayson K., et al. Work Productivity and Activity Impairment Associated with Generalized Anxiety Disorder Among Adults in the United States. presented at: ISPOR 2024; 2024; Atlanta, Georgia.
- [13] Ruscio AM, Hallion LS, Lim CCW, et al. Cross-sectional comparison of the epidemiology of DSM-5 generalized anxiety disorder across the globe. *JAMA Psychiatry* 2017;74(5):465–75. <https://doi.org/10.1001/jamapsychiatry.2017.0056>.
- [14] Hoffman DL, Dukes EM, Wittchen HU. Human and economic burden of generalized anxiety disorder. *Depress Anxiety* 2008;25(1):72–90. <https://doi.org/10.1002/da.20257>.
- [15] Kertz SJ, Woodruff-Borden J. Human and economic burden of GAD, subthreshold GAD, and worry in a primary care sample. *J Clin Psychol Med Settings* 2011;18(3):281–90. <https://doi.org/10.1007/s10880-011-9248-1>.
- [16] Remes O, Wainwright N, Surtees P, et al. Generalised anxiety disorder and hospital admissions: findings from a large, population cohort study. *BMJ Open* 2018;8(10):e018539. <https://doi.org/10.1136/bmjopen-2017-018539>.
- [17] Kavelaars R, Ward H, Mackie DS, et al. The burden of anxiety among a nationally representative US adult population. *J Affect Disord* 2023;336:81–91. <https://doi.org/10.1016/j.jad.2023.04.069>.
- [18] Ferries E, Suponcic S, Chung S, et al. SA5 systematic literature review of unmet needs and burden of disease in generalized anxiety disorder (GAD). *Value Health* 2025;28(6):S395–6. <https://doi.org/10.1016/j.jval.2025.04.1761>.
- [19] Kessler RC, Berglund P, Demler O, et al. Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the national comorbidity survey replication. *Arch Gen Psychiatry* 2005;62(6):593–602. <https://doi.org/10.1001/archpsyc.62.6.593>.
- [20] Spitzer RL, Kroenke K, Williams JB, et al. A brief measure for assessing generalized anxiety disorder: the GAD-7. *Arch Intern Med* 2006;166(10):1092–7. <https://doi.org/10.1001/archinte.166.10.1092>.
- [21] Terlizzi EP, Villarreal MA. Symptoms of generalized anxiety disorder among adults: United States, 2019. *NCHS Data Brief* 2020;(378):1–8.
- [22] Zablotsky B, Black LI, Terlizzi EP, et al. Anxiety and depression symptoms among children before and during the COVID-19 pandemic. *Ann Epidemiol* 2022;75:53–6. <https://doi.org/10.1016/j.annepidem.2022.09.003>.
- [23] U.S. Department of Health and Human Services, Food and Drug Administration, Center for Drug Evaluation and Research, et al. *Best Practices for Conducting and Reporting Pharmacoeconomic Safety Studies Using Electronic Healthcare Data*. 2013.
- [24] Emerson SD, McLinden T, Sereda P, et al. Secondary use of routinely collected administrative health data for epidemiologic research: answering research questions using data collected for a different purpose. *Int J Popul Data Sci* 2024;9(1):2407. <https://doi.org/10.23889/ijpds.v9i1.2407>.
- [25] Bureau U.S.C.. Data from: HHI-02. Health Insurance Coverage Status and Type of Coverage—All Persons by Age and Sex: 2017–2024. 2025.
- [26] Munshi N, Mehra M, van de Velde H, et al. Use of a claims database to characterize and estimate the incidence rate for Castleman disease. *Leuk Lymphoma* 2015;56(5):1252–60. <https://doi.org/10.3109/10428194.2014.953145>.
- [27] Jegede O, Stefanovics EA, Rhee TG, et al. Multimorbidity and correlates of comorbid depression and generalized anxiety disorder in a nationally representative US sample. *J Nerv Ment Dis* 2023;211(5):355–61. <https://doi.org/10.1097/nmd.0000000000001625>.
- [28] Force UPST. Screening for Anxiety Disorders in Adults: US Preventive Services Task Force Recommendation Statement. *JAMA* 2023;329(24):2163–70. <https://doi.org/10.1001/jama.2023.9301>.
- [29] Essadek A, Guenoun T, Gressier F, et al. Post-pandemic changes in anxiety and depression symptom networks among socioeconomically disadvantaged young Adults: A repeated cross-sectional study. *SSM Popul Health* 2025;31:101854. <https://doi.org/10.1016/j.ssmph.2025.101854>.
- [30] Organization WH. *Transforming mental health for all*. World Ment Health Rep 2022.
- [31] Momin A, Rodrigues K, Stead T, et al. The prevalence of undiagnosed anxiety: a national survey. *J Affect Disord Rep* 2023;13:100584. <https://doi.org/10.1016/j.jadr.2023.100584>.
- [32] Vermani M, Marcus M, Katzman MA. Rates of detection of mood and anxiety disorders in primary care: a descriptive, cross-sectional study. *Prim Care Companion CNS Disord* 2011;13(2). <https://doi.org/10.4088/PCC.10m01013>.
- [33] Szuhany KL, Simon NM. Anxiety disorders: a review. *Jama* 2022;328(24):2431–45. <https://doi.org/10.1001/jama.2022.22744>.
- [34] Karlin D, Suponcic SJ, Maculaitis MC, et al. Quantifying the Burden of Undiagnosed Generalized Anxiety Disorder in the US. New York, New York: American Psychiatric Association Annual Meeting; 2024.
- [35] Armbricht E, Shah R, Poorman GW, et al. Economic and humanistic burden associated with depression and anxiety among adults with non-communicable

- chronic diseases (NCCDs) in the United States. *J Multidiscip Health* 2021;14: 887–96. <https://doi.org/10.2147/jmdh.S280200>.
- [36] Armbrrecht E, Shah A, Schepman P, et al. Economic and humanistic burden associated with noncommunicable diseases among adults with depression and anxiety in the United States. *J Med Econ* 2020;23(9):1032–42. <https://doi.org/10.1080/13696998.2020.1776297>.
- [37] Duong P, Mojtabei R, Ferries E, et al. Health-related quality of life and economic burden associated with likely and diagnosed generalized anxiety disorder among the US general adult population. *J Affect Disord* 2026;400:120972. <https://doi.org/10.1016/j.jad.2025.120972>.
- [38] Regier DA, Narrow WE, Clarke DE, et al. DSM-5 field trials in the United States and Canada, Part II: test-retest reliability of selected categorical diagnoses. *Am J Psychiatry* 2013;170(1):59–70. <https://doi.org/10.1176/appi.ajp.2012.12070999%M.23111466>.
- [39] Macdonald-Gagnon G, Stefanovics EA, Potenza MN, et al. Generalized anxiety and mild anxiety symptoms in U.S. military veterans: prevalence, characteristics, and functioning. *J Psychiatr Res* 2024;171:263–70. <https://doi.org/10.1016/j.jpsychires.2024.02.013>.
- [40] Niermann HCM, Voss C, Pieper L, et al. Anxiety disorders among adolescents and young adults: prevalence and mental health care service utilization in a regional epidemiological study in Germany. *J Anxiety Disord* 2021;83:102453. <https://doi.org/10.1016/j.janxdis.2021.102453>.
- [41] Miloyan B, Byrne GJ, Pachana NA. Age-related changes in generalized anxiety disorder symptoms. *Int Psychogeriatr* 2014;26(4):565–72. <https://doi.org/10.1017/s1041610213002470>.